Project Plan: Montgomery County Bus Access

Data

Current RideOn GTFS files and WMATA GTFS API (includes detailed information on bus routes, trips, arrival times, and schedules, paired with data for use in GIS), 2024 5-year estimate Census data at the tract level from the American Community Survey: Age, sex, household size, number of workers in a household, and number of vehicles available. Household size and number of workers are each separately paired with number of vehicles available. Additional state/county GIS data: hospitals in Maryland, property polygons and other shapefiles/geodatabases from MNCPPC - Montgomery County Data Catalog. These are for comparing location of bus routes to buildings.

Research goals and questions

Analyze existing bus routes in Montgomery County compared to Census data and determine neighborhoods that indicate a need for greater bus coverage or improvement to existing routes. Determine areas that need busses to upcoming Flash BRT stations or existing stations, and determine regions that would benefit the most from greater frequency in existing routes.

1. **What cities and neighborhoods in the county indicate the greatest need for bus coverage? Do the bus routes in place reflect this need?**
    - ArcGIS + R: Examine Census data on number of vehicles available to a household + number of workers in the household.
    - Determine if bus routes are reflecting need using linear regression, predicting number of routes by percentage of households with less than 2 vehicles.
2. **How useful are existing neighborhood bus routes?**
    - R + ArcGIS: Create "neighborhood bus routes" using spatial joins in ArcGIS (routes that terminate/start within a neighborhood), examine bus timetables ("trips", "stop times", "routes" in GTFS) using R.
    - R (+ ArcGIS): Use classification to examine strength of correlation between route service time and whether or not it terminates in a neighborhood.
3. **How many residents live in cities with effective bus coverage? Which cities have the greatest and poorest bus coverage?**
    - ArcGIS (and R): Calculate "effective bus coverage" based on amount of bus routes in region; number of homes ½ mile from a bus stop; bus frequency; number of routes connecting neighborhoods to transfer stations. Compare to Census data on population.
4. **Are proposed Flash BRT stations accessible to residents of the cities they're going to be built in?**

- ArcGIS: Map proposed Flash BRT stations and compare to neighborhood bus routes, based on created efficiency metrics.

5. **What is the average probability of catching a transfer? Where and which routes score the lowest?**
    - R: Calculate probability of arriving at each bus stop along a route soon enough to catch the bus (~5mins before arrival) based on "stop times" (GTFS)
6. **Which cities are the hardest to maneuver through with existing bus routes?**
    - R + ArcGIS: Amount of routes running at the same time in a city; compare to "efficiency" and transfer probability, amount of land in city covered by routes.
7. **What estimated number of residents live within ½ mile of a bus stop?**
    - ArcGIS (+ R): Get total population on Census estimates; use spatial joins to determine if there's a bus stop within ½ mile of each home in a census tract.
8. **How many hospitals are accessible by bus? Do the hospitals nearest to people without vehicles have effective bus coverage?**
    - ArcGIS (+ R): Look at bus routes that pass/end at hospitals, and when they run; look at hospitals near areas with many households without vehicles.

Risks and mitigations

I may face difficulty in calculating bus route access from homes. I haven't found data on the number of people living in each residential home, so I will have to estimate. Additionally, estimating the number of residents within ½ mile of a bus stop might not be possible with the available data; I might need to use the number of residential homes within that distance instead. I will also need to do a lot of research on transit planning.

Milestones